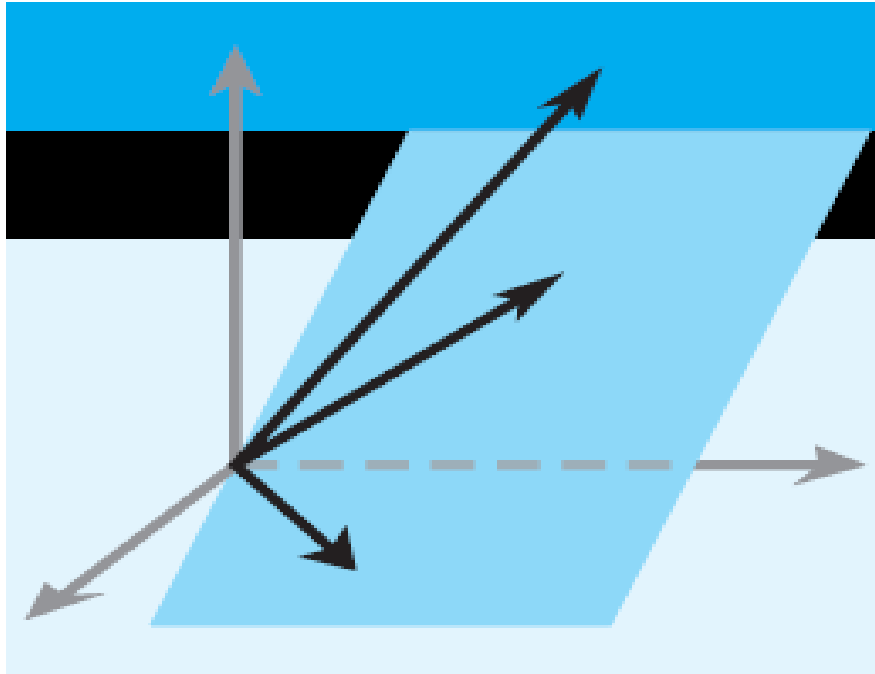


Elementary Linear Algebra

Chapter 6



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Chapter 6 Inner Product Spaces

- 6.1 Inner Products
- 6.2 Angle and Orthogonality in Inner Product Spaces
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Section 6.4 Best Approximation; Least Squares



THEOREM 6.4.1 Best Approximation Theorem Page 360

*If W is a finite-dimensional subspace of an inner product space V , and if $\mathbf{b}$ is a vector in V , then $\text{proj}_W \mathbf{b}$ is the **best approximation** to $\mathbf{b}$ from W in the sense that*

$$\|\mathbf{b} - \text{proj}_W \mathbf{b}\| < \|\mathbf{b} - \mathbf{w}\|$$

for every vector $\mathbf{w}$ in W that is different from $\text{proj}_W \mathbf{b}$.

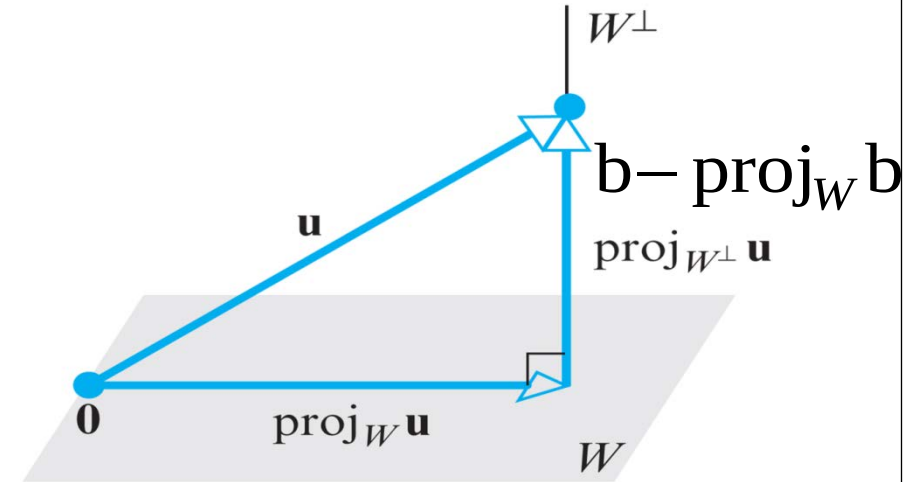
Section 6.4 Best Approximation; Least Squares

Proof: We want to show $\|b - w\| > \|b - \text{proj}_W b\|$

$$\|b - w\|$$

$$\| \underline{b - \text{proj}_W b} + \underline{\text{proj}_W b - w} \|$$

$$\text{proj}_W b \in W \quad w \in W$$



➡ $\langle b - \text{proj}_W b, \text{proj}_W b \rangle = ?$

➡ $\langle b - \text{proj}_W b, \text{proj}_W b - w \rangle = ?$

by orthogonal

➡ $\|b - w\| = \| \underline{b - \text{proj}_W b} \| + \| \underline{\text{proj}_W b - w} \|$

*by theorem of
Pythagoras*

➡ $\|b - w\| > \|b - \text{proj}_W b\|$

Section 6.4 Best Approximation; Least Squares

Section 4.8 Rank, Nullity, and the Fundamental Matrix Spaces

THEOREM 4.8.6 Let A be an $m \times n$ matrix. **Page 218**



(a) (**Overdetermined Case**). If $m > n$, then the linear system $A\mathbf{x} = \mathbf{b}$ is inconsistent for at least one vector $\mathbf{b}$ in $\mathbf{R}^n$

(a) (**Underdetermined Case**). If $m < n$, then for each vector $\mathbf{b}$ in $\mathbf{R}^m$ the linear system $A\mathbf{x} = \mathbf{b}$ is either inconsistent or has infinitely many solutions.

Section 6.4 Best Approximation; Least Squares

▶ EXAMPLE 1

(a) Find solutions of the linear system

$$A = \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$$
$$\begin{array}{rclcl} x_1 & - & x_2 & = & 4 \\ 3x_1 & + & 2x_2 & = & 1 \\ -2x_1 & + & 4x_2 & = & 3 \end{array}$$

Rank of A =
Nullity of A =

$$\begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} \xrightarrow{\text{Row Operations}} \begin{bmatrix} 1 & -1 & 4 \\ 0 & 5 & -11 \\ 0 & 0 & 33/5 \end{bmatrix} \xrightarrow{\text{inconsistent}}$$

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THEOREM 4.7.1 *A system of linear equations $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A .*

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$A\mathbf{x} = \boxed{\phantom{a_{11}x_1 + a_{21}x_2 + \cdots + a_{m1}x_n}} = \boxed{\phantom{a_{11}x_1}} + \boxed{\phantom{a_{21}x_2}} + \cdots + \boxed{\phantom{a_{m1}x_n}} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

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$b \in \text{col}(A) \implies Ax = b$ is consistent

$b \notin \text{col}(A) \implies Ax = b$ is inconsistent

Idea: We would like to find $b^ \in \text{col}(A)$ such that $Ax = b^*$
 $Ax^* \approx b$*

$b \in R^m$ $W = \text{col}(A)$

We want b^ close to b , namely $\|b - b^*\|$ is smallest.*

Based on best approximated theorem $\text{proj}_W b \in W$

$Ax = \text{proj}_W b \implies x = ?$

$\implies b - Ax = b - \text{proj}_W b$

Question: $b - \text{proj}_W b \in ?$ $W^\perp$ $\langle b - \text{proj}_W b, ? \rangle = 0$

$W = \text{col}(A)$

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➡ $A^T (b - Ax) = A^T (b - \text{proj}_W b)$

➡ $A^T (b - Ax) = A^T (b - \text{proj}_W b)$

➡ $A^T (b - Ax) =$ 

orthogonal complement

➡ $A^T Ax = Ab$

Matrix multiplication

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THEOREM 6.4.2 *For every linear system $A\mathbf{x} = \mathbf{b}$, the associated normal system*

$$\underline{A^T A \mathbf{x} = A^T \mathbf{b}} \quad (5)$$

is consistent, and all solutions of (5) are least squares solutions of $A\mathbf{x} = \mathbf{b}$. Moreover, if W is the column space of A , and $\mathbf{x}$ is any least squares solution of $A\mathbf{x} = \mathbf{b}$, then the orthogonal projection of $\mathbf{b}$ on W is

$$\text{proj}_W \mathbf{b} = A\mathbf{x} \quad (6)$$

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▶ EXAMPLE 1 : Least Squares Solution

(a) Find all least squares solutions of the linear system

$$\begin{array}{rcl} x_1 & - & x_2 = 4 \\ 3x_1 & + & 2x_2 = 1 \\ -2x_1 & + & 4x_2 = 3 \end{array}$$

(b) Find the error vector and the error.

Solution (a) It will be convenient to express the system in the matrix form $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix}$$

Section 6.4 Best Approximation; Least Squares

► EXAMPLE 1

Solution (a)

It follows that

$$A^T A = \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 14 & -3 \\ -3 & 21 \end{bmatrix}$$

$$A^T \mathbf{b} = \begin{bmatrix} 1 & 3 & -2 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

Section 6.4 Best Approximation; Least Squares

▶ EXAMPLE 1

Solution (a)

so the normal system $A^T A \mathbf{x} = A^T \mathbf{b}$ is

$$\begin{bmatrix} 14 & -3 \\ -3 & 21 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

Solving this system yields a unique least squares solution, namely,

$$x_1 = \frac{17}{95}, \quad x_2 = \frac{143}{285}$$

Section 6.4 Best Approximation; Least Squares

► EXAMPLE 1

Solution (*b*) The error vector is

$$\mathbf{b} - A\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} \frac{17}{95} \\ \frac{143}{285} \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} -\frac{92}{285} \\ \frac{439}{285} \\ \frac{95}{57} \end{bmatrix} = \begin{bmatrix} \frac{1232}{285} \\ -\frac{154}{285} \\ \frac{4}{3} \end{bmatrix}$$

$$\|\mathbf{b} - A\mathbf{x}\| \approx 4.556$$

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► EXAMPLE 2 Orthogonal Projection on a Subspace

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Find the orthogonal projection of the vector $\mathbf{u} = (-3, -3, 8, 9)$ on the subspace of $\mathbf{R}^4$ spanned by the vectors

$$\mathbf{u}_1 = (3, 1, 0, 1), \quad \mathbf{u}_2 = (1, 2, 1, 1), \quad \mathbf{u}_3 = (-1, 0, 2, -1)$$

Solution

Method 1. We could solve this problem by first using the Gram-Schmidt process to convert $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ into an orthonormal basis and then applying the method used in Example 6 of Section 6.3.

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► EXAMPLE 2 Orthogonal Projection on a Subspace

Solution : Method 2.

However, the following method is more efficient.

The subspace W of $\mathbf{R}^4$ spanned by $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ is the column space of the matrix

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \\ 1 & 1 & -1 \end{bmatrix}$$

The subspace W of $\mathbf{R}^4$ spanned by $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ is the column space of the matrix

Section 6.4 Best Approximation; Least Squares

► EXAMPLE 2 Orthogonal Projection on a Subspace

Thus, if $\mathbf{u}$ is expressed as a column vector, we can find the orthogonal projection of $\mathbf{u}$ on W by finding a least squares solution. The computations are as follows:

The system $A\mathbf{x} = \mathbf{u}$ is

$$\begin{bmatrix} 3 & 1 & -1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \\ 8 \\ 9 \end{bmatrix}$$

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► EXAMPLE 2

So

$$A^T A = \begin{bmatrix} 3 & 1 & 0 & 1 \\ 1 & 2 & 1 & 1 \\ -1 & 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 11 & 6 & -4 \\ 6 & 7 & 0 \\ -4 & 0 & 6 \end{bmatrix}$$

$$A^T \mathbf{u} = \begin{bmatrix} 3 & 1 & 0 & 1 \\ 1 & 2 & 1 & 1 \\ -1 & 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} -3 \\ -3 \\ 8 \\ 9 \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ 10 \end{bmatrix}$$

The normal system $A^T A \mathbf{x} = A^T \mathbf{u}$ in this case is

$$\begin{bmatrix} 11 & 6 & -4 \\ 6 & 7 & 0 \\ -4 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 8 \\ 10 \end{bmatrix}$$

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► EXAMPLE 2 Orthogonal Projection on a Subspace

Solving this system yields

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

as the least squares solution of $A\mathbf{x} = \mathbf{u}$ (verify), so

$$\text{proj}_W \mathbf{u} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ 4 \\ 0 \end{bmatrix}$$

or, in comma-delimited notation, $\text{proj}_W \mathbf{u} = (-2, 3, 4, 0)$.

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THEOREM 6.4.3 Page 363

If A is an $m \times n$ matrix, then the following are equivalent.

(a) The column vectors of A are linearly independent.

(b) $A^T A$ is invertible.

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THEOREM 6.4.4

If A is an $m \times n$ matrix with linearly independent column vectors, then for every $m \times 1$ matrix $\mathbf{b}$, the linear system $A\mathbf{x} = \mathbf{b}$ has a unique least squares solution. This solution is given by

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b} \quad (9)$$

Moreover, if W is the column space of A , then the orthogonal projection of $\mathbf{b}$ on W is

$$\text{proj}_W \mathbf{b} = A\mathbf{x} = A (A^T A)^{-1} A^T \mathbf{b} \quad (10)$$

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THEOREM 6.4⁵

If A is an $m \times n$ matrix with linearly independent column vectors, and if $A = QR$ is a QR -decomposition of A (see Theorem [6.3.7](#)), then for each $\mathbf{b}$ in $\mathbb{R}^m$ the system $A\mathbf{x} = \mathbf{b}$ has a unique least squares solution given by

$$\mathbf{x} = R^{-1}Q^T\mathbf{b} \quad (9)$$

DEFINITION 1 If W is a subspace of $\mathbb{R}^m$, then the linear transformation $P: \mathbb{R}^m \rightarrow W$ that maps each vector $\mathbf{x}$ in $\mathbb{R}^m$ into its orthogonal projection $\text{proj}_W \mathbf{x}$ in W is called the **orthogonal projection of $\mathbb{R}^m$ on W** .

► EXAMPLE 3 The Standard Matrix for an Orthogonal Projection on a Line

We showed in Formula (16) of Section 4.9 that

$$P_{\theta} = \begin{bmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{bmatrix}$$

Is the standard matrix for the orthogonal projection on the line W through the origin of $\mathbf{R}^2$ that makes an angle θ with the positive x -axis. Derive this result using Formula (10).

Solution The column vectors of A can be formed from any basis for W . Since W is one-dimensional, we can take $\mathbf{w} = (\cos \theta, \sin \theta)$ as the basis vector (Figure 6.4.2), so

$$A = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

We leave it for you to show that $A^T A$ is 1×1 identity matrix. Thus, Formula (10) simplifies to

$$\begin{aligned} A(A^T A)^{-1} A^T &= A A^T = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} [\cos \theta \ \sin \theta] \\ &= \begin{bmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{bmatrix} = P_{\theta} \end{aligned}$$

THEOREM 6.4.6 Equivalent Statements

If A is an $n \times n$ matrix, then the following statements are equivalent.

- (a) A is invertible.
- (b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (c) The reduced row echelon form of A is I_n .
- (d) A is expressible as a product of elementary matrices.
- (e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.
- (f) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $n \times 1$ matrix $\mathbf{b}$.
- (g) $\det(A) \neq 0$.
- (h) The column vectors of A are linearly independent.
- (i) The row vectors of A are linearly independent.
- (j) The column vectors of A span R^n .
- (k) The row vectors of A span R^n .
- (l) The column vectors of A form a basis for R^n .
- (m) The row vectors of A form a basis for R^n .
- (n) A has rank n .
- (o) A has nullity 0.
- (p) The orthogonal complement of the null space of A is R^n .
- (q) The orthogonal complement of the row space of A is $\{\mathbf{0}\}$.
- (r) The range of T_A is R^n .
- (s) T_A is one-to-one.
- (t) $\lambda = 0$ is not an eigenvalue of A .
- (u) $A^T A$ is invertible.

Homework

Sec 6.4

True/False

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98, 100(b), 102(b)

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